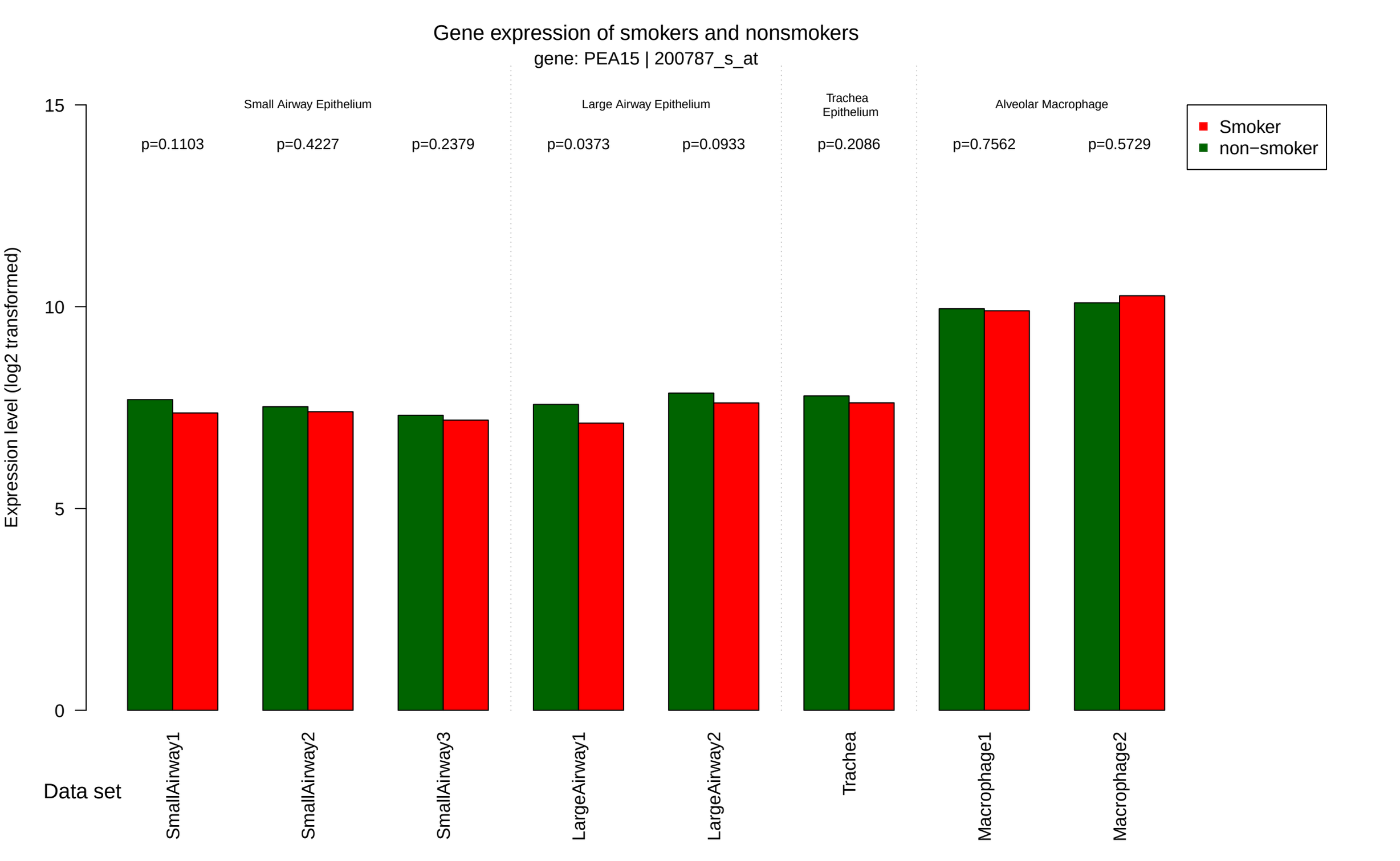
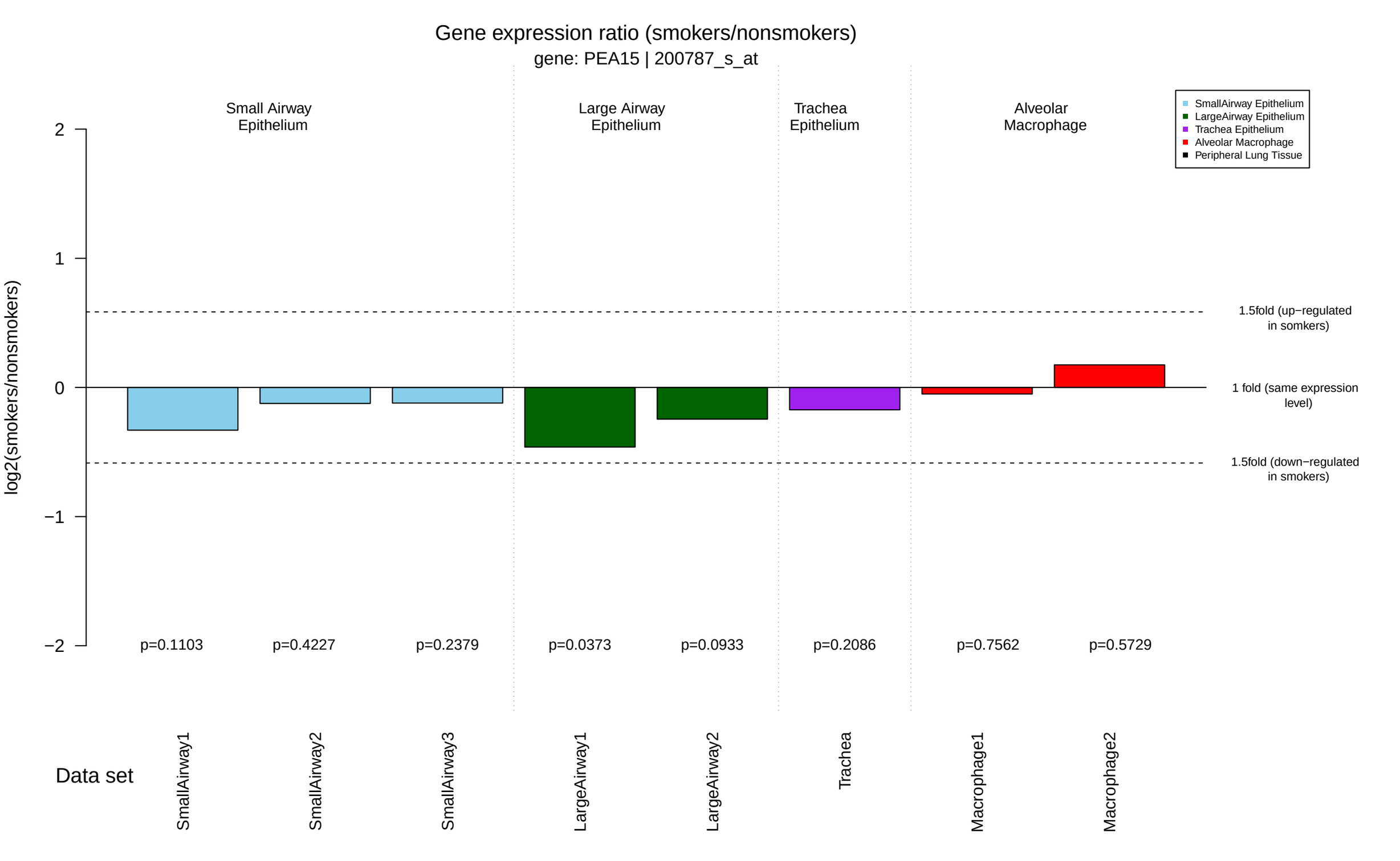
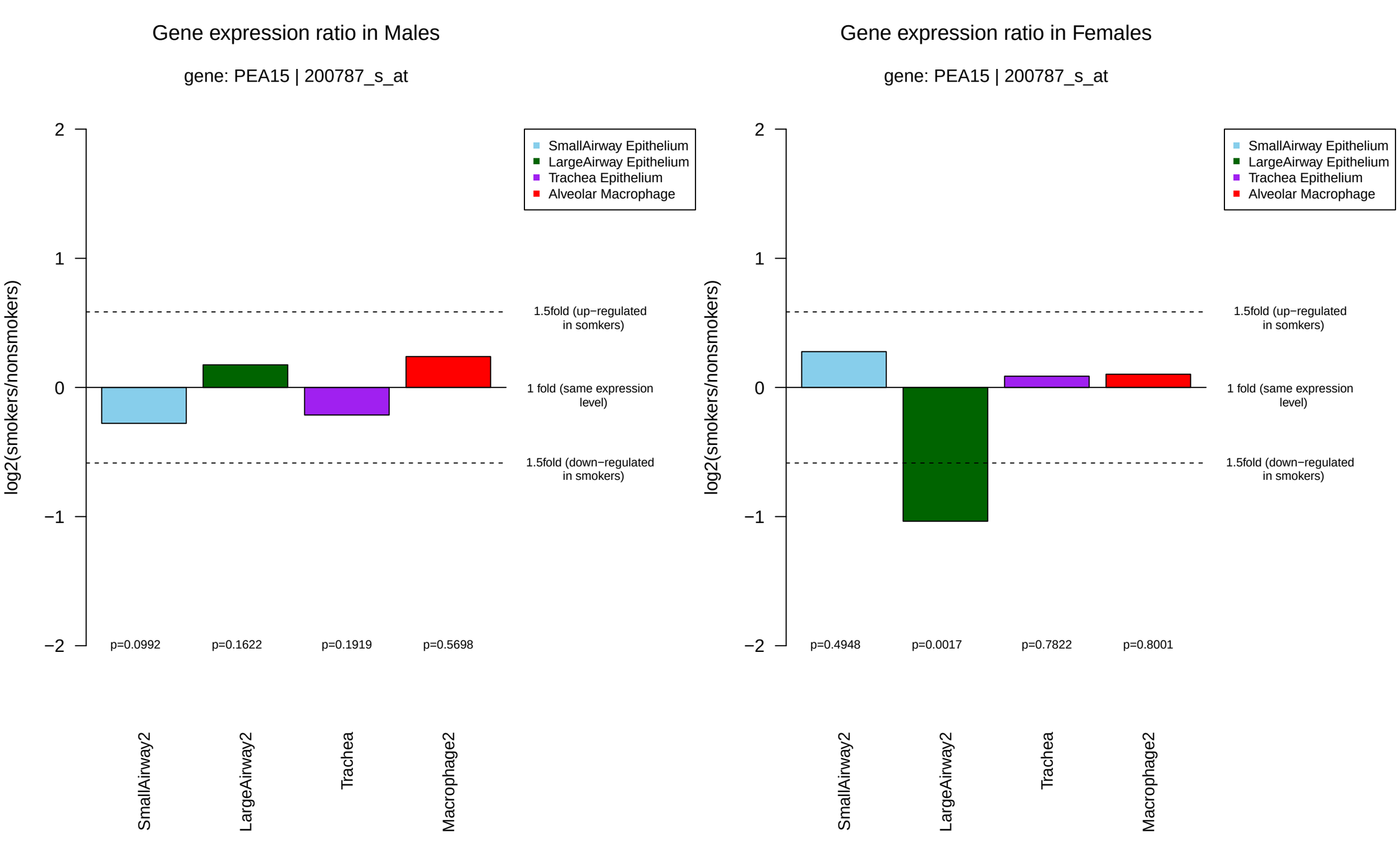
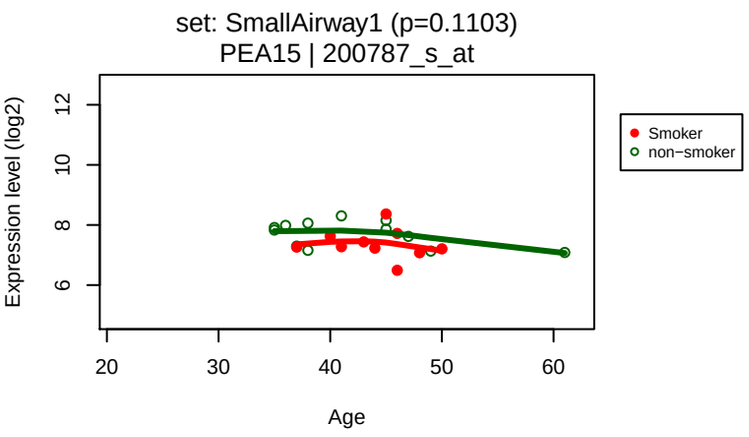
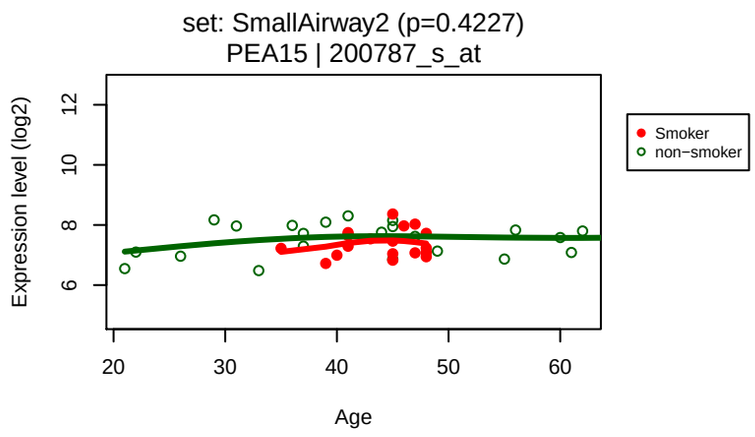
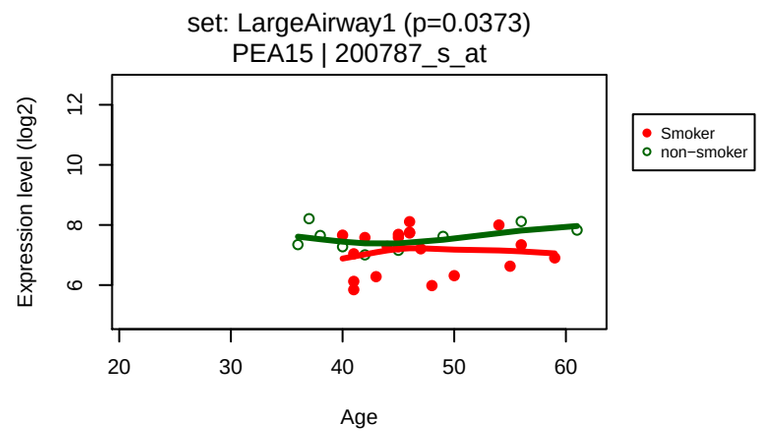

Gene expression of smokers and nonsmokers
gene: PEA15 | 200787_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1103
p=0.4227
p=0.2379
p=0.0373
p=0.0933
p=0.2086
p=0.7562
p=0.5729
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

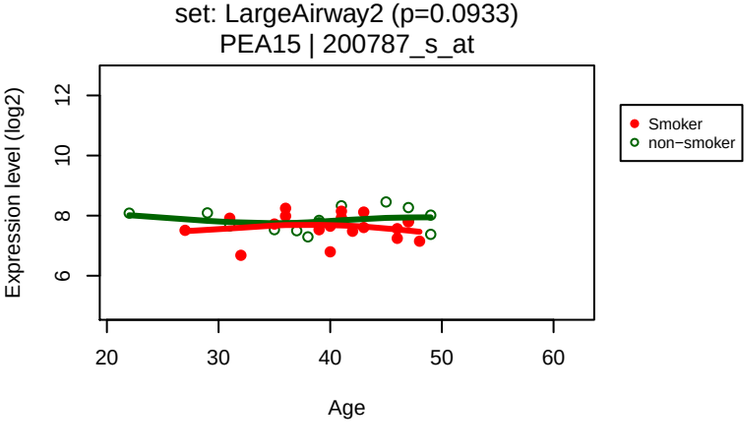

Gene expression ratio (smokers/nonsmokers)
gene: PEA15 | 200787_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1103
p=0.4227
p=0.2379
p=0.0373
p=0.0933
p=0.2086
p=0.7562
p=0.5729
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

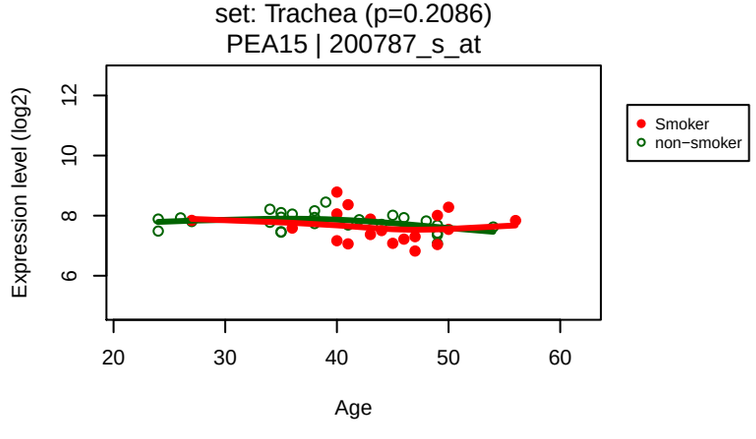

Gene expression ratio in Males
gene: PEA15 | 200787_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0992
p=0.1622
p=0.1919
p=0.5698
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEA15 | 200787_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4948
p=0.0017
p=0.7822
p=0.8001
SmallAirway2
LargeAirway2
Trachea
Macrophage2

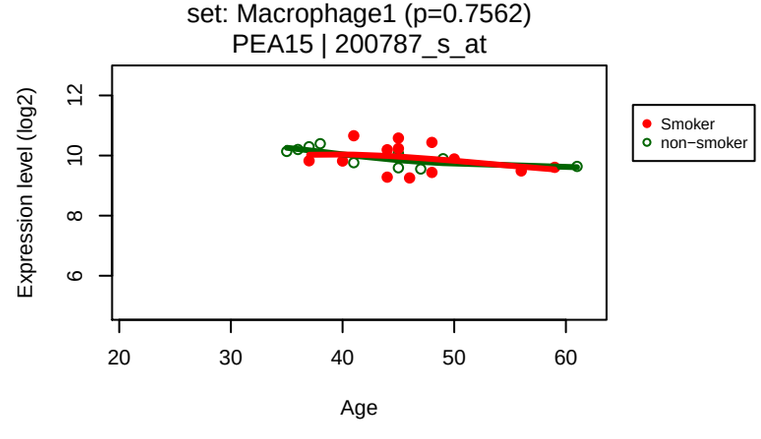

set: SmallAirway1 (p=0.1103)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker

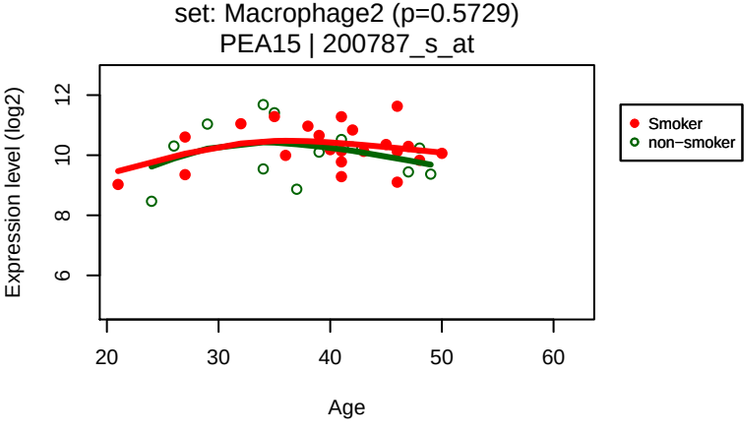

set: SmallAirway2 (p=0.4227)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker

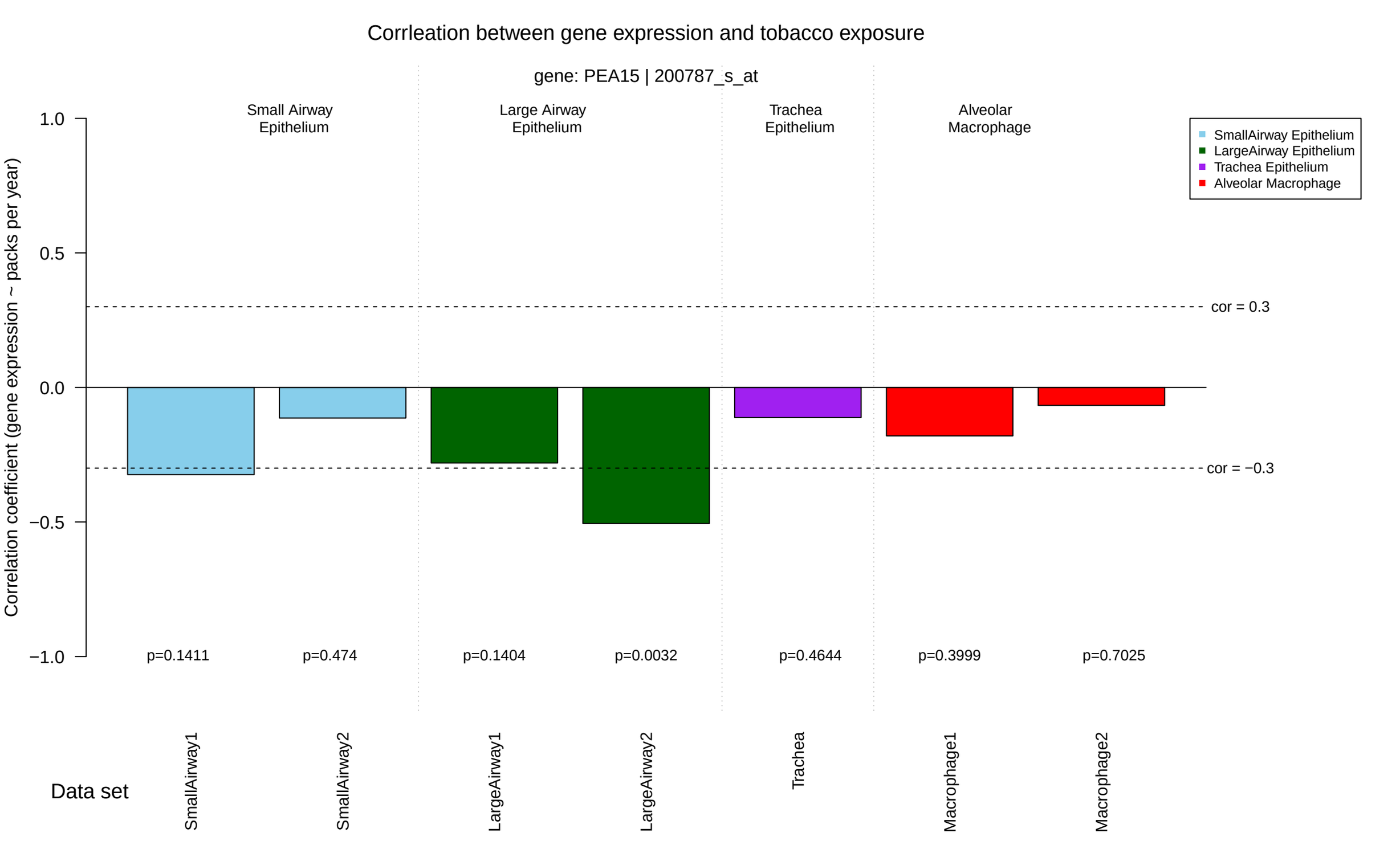

set: LargeAirway1 (p=0.0373)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0933)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2086)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7562)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5729)
PEA15 | 200787_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEA15 | 200787_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1411
p=0.474
p=0.1404
p=0.0032
p=0.4644
p=0.3999
p=0.7025
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2